

1. Abstract and Environmental Parameters

This study analyzes the combined thermal and electrical conversion efficiency of dual-axis tracking heliostat arrays under the standard solar irradiance model AM1.5D (Direct solar irradiance 1000 W/m² at 25C).

2. Thermal Receiver Efficiency

The molten salt central receiver attained a peak thermal conversion efficiency of 41.2% at a working temperature of 650C. Receiver heat loss was mitigated using high-emissivity ceramic coatings.

3. Molten Salt Thermal Energy Storage (TES)

The two-tank indirect molten salt thermal storage system provides up to 8.0 hours of full-load turbine generation buffer, enabling high capacity factor operation during non-daylight hours.

Figure 2: Daily Power Output Profile with Thermal Storage

The chart illustrates baseline solar generation peaking at 13:00 (120 MWth) followed by molten salt dispatch sustaining 85 MWe electrical output from 17:00 to 01:00. Overall plant capacity factor increased from 27.5% to 58.4%.